RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

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Academic year 2024-2025 (EVEN Sem)

EPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	02nd April 2025	Maximum Marks	10 + 50
Course Code	IM244AI	Duration	30 + 90 Min
Sem	IV	CIE – I	

OPERATIONS RESEARCH

NOTE:
1. Answer all the Questions.

Sl. No.	Questions	M	BT	CO
	Part – A			
1	The variables that represent decisions to be made in an optimization model are called _____.	1	1	1
2	In an optimization model, the function that needs to be maximized or minimized is called the _____.	1	1	1
3	Linear Programming is a mathematical technique used for _____ decision-making.	1	2	2
4	One of the key limitations of Operations Research is that it requires _____ data for accuracy.	1	2	2
5	The feasible region is the set of all possible solutions that satisfy all _____.	1	1	1
6	The graphical solution method of LPP is applicable only when there are _____ decision variables.	1	2	2
7	The additivity assumption ensures that the total effect of different decision variables is the _____ of their individual effects.	1	2	1
8	When the total supply does not equal total demand, the transportation problem is called a(n) _____ transportation problem.	1	2	2
9	A prohibited route transportation problem occurs when certain routes have _____ restrictions and cannot be used for transportation.	1	2	3
10	The General Mathematical Formulation of a transportation problem consists of an objective function, supply constraints, and _____ constraints.	1	2	2
	Part –B			
1	Describe the Seven-Step Model-Building Process in Operations Research with an example.	10	2	1
2	Explain the key features of Operations Research (OR) models and highlight its limitations.	10	2	1
3	Consider the following problem faced by a production planner of a soft drink plant. He has two bottling machines A and B. A is designed for 8-ounce bottles and B for 16-ounce bottles. However, each can also be used for both types of bottles with some loss of efficiency. The manufacturing data is as follows: **Machine** | **8-ounce Bottles** | **16-ounce Bottles** A | 100/minute | 40/minute B | 60/minute | 75/minute The machines can be run for 8 hours per day, 5 days per week. The profit on an	10	3	2

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	8-ounce bottle is Rs 1.5 and on a 16-ounce bottle is Rs 2.5. Weekly production of the drink cannot exceed 3,00,000 bottles and the market can absorb 25,000, 8-ounce bottles and 7,000, 16-ounce bottles per week. The planner wishes to maximize his profit, subject of course, to all the production and marketing restrictions. Formulate this problem as an LP model to maximize total profit.			
4	A diet for a sick person must contain at least 4,000 units of vitamins, 50 units of minerals and 1,400 calories. Two foods A and B are available at a cost of Rs. 4 and Rs. 3 per unit, respectively. If one of A contains 200 units of vitamins, 1 unit of mineral and 40 calories and one unit of food B contains 100 units of vitamins, 2 units of minerals and 40 calories. Formulate this problem as an LP model and solve it by graphical method to find combination of foods to be used to have least cost.	10	3	2
5	Determine an initial basic feasible solution to the following transportation problem by using (a) NWCR, (b) LCM and (c) VAM.	10	3	3

	D1	D2	D3	D4	Supply
S1	19	30	50	10	7
S2	70	30	40	60	9
S3	40	8	70	20	18
Demand	5	8	7	14	

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Quiz	Max Marks	4	5	1	-	3	7	-	-	-	-
	Test		20	20	10	-	-	20	30	-	-	-

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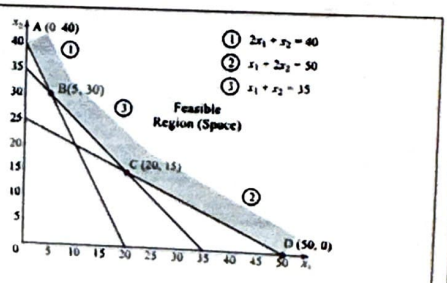
Academic year 2024-2025 (ODD Sem)

EPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**02/04/2025**	**Maximum Marks**	**10 + 50**
Course Code	**IM244AI**	**Duration**	**30 + 90 Min**
Sem	**IV**	**CIE – I-Scheme**	
Operations Research			

Note:
1. Answer all the Questions.

Sl. No.	**Questions**	**M**
	Part – A	
1	decision variables	1
2	objective function	1
3	optimization	1
4	accurate	1
5	constraints	1
6	two	1
7	sum	1
8	unbalanced	1
9	zero/infinite cost	1
10	demand	1
	Part –B	
1	The Seven-Step Model-Building Process is used to structure and solve decision-making problems systematically. 1. Problem Definition 2. Data Collection 3. Model Formulation 4. Model Solution 5. Model Validation 6. Implementation 7. Monitoring & Updating Listing-3M,Explanation-7M	10
2	**Features of OR Models:** Scientific Approach, Optimization Focus ,Interdisciplinary Nature ,Quantitative Decision-Making, Systems Approach, Simulation & Prediction , Iterative Process ---5M **Limitations of OR**: Complexity, Data Dependency ,High Computational Requirements, Time-Consuming ,Assumptions vs. Reality, Implementation Challenges, Not Always Practical--------5M	10

3	**LP model formulation** The data of the problem is summarized as follows: | Constraints	Production		Availability | |---|---|---|---| |	8-ounce Bottles	16-ounce Bottles	| | Machine A time	100/minute	40/minute	$8 \times 5 \times 60 = 2{,}400$ minutes | | Machine B time	60/minute	75/minute	$8 \times 5 \times 60 = 2{,}400$ minutes | | Production	1	1	3,00,000 units/week | | Marketing	1	–	25,000 units/week | |	–	1	7,000 units/week | | Profit/unit (Rs)	1.5	2.5	| ***Decision variables*** Let x_1 and x_2 = units of 8-ounce and 16-ounce bottles to be produced weekly, respectively ***The LP model*** Maximize (total profit) $Z = 1.5x_1 + 2.5x_2$ **subject to the constraints** (i) Machine time : (a) $\frac{x_1}{100} + \frac{x_2}{40} \le 2{,}400$ and (b) $\frac{x_1}{60} + \frac{x_2}{75} \le 2{,}400$ (ii) Production : $x_1 + x_2 \le 3{,}00{,}000$ (iii) Marketing : (a) $x_1 \le 25{,}000$, (b) $x_2 \le 7{,}000$ **and** $x_1, x_2 \ge 0$. Table(4M)+decision variables and objective(2M)+constraints(4M)	10
4	**Solution** The data of the given problem can be summarized as shown below. | Food | Units Content of | | | Cost per Unit (Rs.) | |---|---|---|---|---| | | Vitamins | Mineral | Calories | | | A | 200 | 1 | 40 | 4 | | B | 100 | 2 | 40 | 3 | | Minimum requirement | 4,000 | 50 | 1,400 | | Let us define the following decision variables: x_1 and x_2 = number of units of food A and B to be used, respectively. Then LP model of the given problem is: Minimize (total cost) $Z = 4x_1 + 3x_2$ subject to the constraints (i) Vitamins : $200x_1 + 100x_2 \ge 4{,}000$, (ii) Mineral : $x_1 + 2x_2 \ge 50$, (iii) Calories : $40x_1 + 40x_2 \ge 1{,}400$ and $x_1, x_2 \ge 0$ Formulation-4M  Graph-3M,Answer-3M	10
5	NWC:Total cost = $5 \times 19 + 2 \times 30 + 6 \times 30 + 3 \times 40 + 4 \times 70 + 14 \times 20$ = Rs 1,015------3M LCM:Total cost = $7 \times 10 + 2 \times 70 + 7 \times 40 + 3 \times 40 + 8 \times 8 + 7 \times 20$ = Rs 814-------3M VAM: Total cost = $5 \times 19 + 2 \times 10 + 7 \times 40 + 2 \times 60 + 8 \times 8 + 10 \times 20$ = Rs 779—4M	10

Academic year 2023-2024 (EVEN Sem)

DEPARTMENT OF

INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**19th June 2024**	**Maximum Marks**	**50**
Course Code	**IM244AI**	**Duration**	**90 Min**
Sem	**IV Semester**	**CIE – I**	
OPERATIONS RESEARCH			

Note:

1. Answer all the Questions.

Sl. No.	Questions	M	BT	CO
1	In relation to the LP problem, explain the implications of the following assumptions of the model: i. Linearity of the objective function and constraints, ii. Continuous variables, iii. Certainty.	10	1	1
2	The company manufactures the models A, B and C whose profit contributions per unit are Rs 20, Rs 50 and Rs 70 respectively. The minimum weekly production requirement is 30 units for Model A, 150 units for Model B and 60 units for Model C. Each type of tape recorder requires a certain amount of time to manufacture the component parts for assembly and packaging. Specifically, dozens of Model A units require 6 hours to manufacture, 5 hours to assemble, and 2 hour to package. The numbers corresponding to a dozen units of Model B are 5, 6 and 4 and for dozens of Model C units are 7, 8 and 5. Over the next week, the company provides 150 hours of manufacturing, 140 hours of assembly and 60 hours of packing time And packaging. Formulate this problem as a LP model for maximizing the company's gross profit.	10	2	2
3	Solve the following Linear Programming Problem by Graphical Method: Maximize $Z = 300x_1 + 400x_2$ Subject to the constraints $5x_1 + 4x_2 \leq 200$ $3x_1 + 5x_2 \leq 150$ $5x_1 + 4x_2 \geq 100$ $8x_1 + 4x_2 \geq 80$ where $x_1, x_2 \geq 0$	10	3	3
4	There are four machines and four operators with their respective payoffs (in 100 rupees) as shown below. How the machines should be allocated to each operator so as to minimize the total pay – off. Operators Machines O_1 O_2 O_3 O_4 M_1 5 4 2 7 M_2 7 8 3 5 M_3 5 3 4 6 M_4 4 6 6 5	10	2	2

5	Five salesmen are to be assigned to five districts. Estimates of sales revenue (in thousands) for each salesman are given as follows: Find the assignment pattern that maximizes the sales revenue.	10	3	3

	A	B	C	D	E
1	32	38	40	28	40
2	40	24	28	21	36
3	41	27	33	30	37
4	22	38	41	36	36
5	29	33	40	35	39

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	10	20	20	–	10	20	20	–	–	–

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